Machine-Learning-Driven Catalyst Screening for the Electrochemical Oxidation of 5-Hydroxymethylfurfural to 2,5-Furandicarboxylic Acid

Research report generated from the HMF-Connected pipeline · 11 November 2026

ABSTRACT: Electrochemical oxidation of 5-hydroxymethylfurfural (HMF) to 2,5-furandicarboxylic acid (FDCA) couples biomass upgrading with hydrogen evolution, but catalyst development still relies on trial-and-error experiments. Following the machine-learning (ML) screening paradigm of Li et al. [1], we assembled 154 electro-oxidation records from 99 publications and encoded 61 descriptors of the catalyst, electrolyte, potential, temperature, TEMPO mediation and cell configuration. Records were grouped by K-means clustering in the conversion-yield space (K = 3-6). Class imbalance was handled with random oversampling, SMOTE and ADASYN applied inside training folds only. Six classifiers were compared under 25 repeats of 5-fold publication-grouped cross-validation, which scores every model on papers it has never seen. The NN model with SMOTE gave the best macro-F1 for the ≥90% FDCA performance screen (0.731; dummy baseline 0.480), and gradient boosting reached 0.626 for K = 3. SHAP analysis linked performance to the electrode support, applied potential and HMF concentration. A validation-gated NSGA-II search with mixed encoding and entropy-weighted decision making, restricted to $Ni(OH)_2$ or $Cu(OH)_2$ anodes, converged on $Ni(OH)_2$ on nickel foam in ~1.1 M KOH with ~11 mM HMF at 1.41 V vs RHE and 25 °C. This region agrees with published $Ni(OH)_2$ electrodes that reach 95-99% FDCA yield.

KEYWORDS: *machine learning, 5-hydroxymethylfurfural, 2,5-furandicarboxylic acid, electrocatalysis, oversampling, SHAP, genetic algorithm, NSGA-II*

1. INTRODUCTION

FDCA is a bio-based replacement for terephthalic acid. Polymerised with ethylene glycol it gives poly(ethylene furanoate), whose life-cycle energy use and greenhouse-gas emissions are lower than those of PET [3]. Electrochemical oxidation of HMF to FDCA runs at ambient temperature and pressure. It uses water as the oxygen source and can be paired with cathodic hydrogen evolution [2], avoiding the pressurised O_2, noble metals and added bases of thermal aerobic routes. The reaction proceeds through HMFCA or DFF and FFCA intermediates. Its rate and selectivity depend jointly on the anode material, electrolyte pH, applied potential and cell design, which makes catalyst selection a high-dimensional problem.

Li et al. [1] showed that this problem can be attacked with a data-model-synthesis loop for the thermocatalytic route. Their loop runs K-means classification of literature data, composite oversampling, a four-model ML comparison, SHAP interpretation and NSGA-II optimisation, and it identified $Ru/Mn_6Ce_1O_Y$ as a promising catalyst. Here we transfer the same workflow to the *electrochemical* route. Three changes matter for honest model assessment on a small literature set:

(i) Validation is **publication-grouped**, so that records from one paper never appear in both training and test folds. (ii) Oversampling happens **only inside training folds**, and repeats in which a sampler is infeasible for any fold are excluded rather than silently scored on a subset. (iii) The genetic algorithm is **gated** on regressors that beat a median baseline, and its search space is bounded by conditions actually reported for the catalyst family of interest.

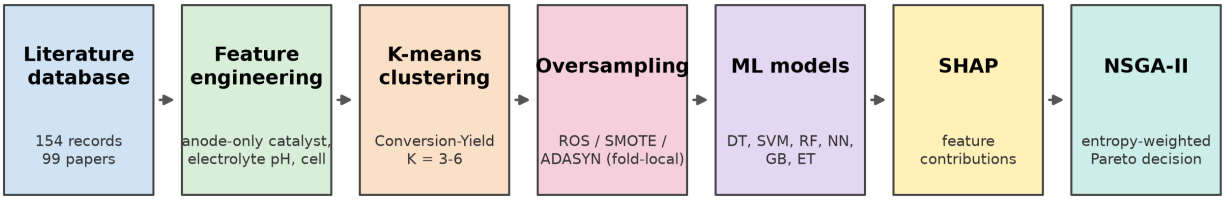


Figure 1. Workflow of the study: literature database, feature engineering, K-means classification, fold-local oversampling, six-model comparison, SHAP interpretation and NSGA-II catalyst/condition optimisation.

2. DATA AND METHODS

2.1. Data Preprocessing.

The database contains 154 records of HMF electro-oxidation extracted from 99 publications. Each record lists the cell type, anode/cathode materials, electrolyte, HMF concentration, potential, TEMPO use, temperature and the measured conversion, FDCA yield, faradaic efficiency (FE) and selectivity, reported for 117, 123, 119 and 91 records, respectively. Free text was parsed into numerical descriptors. Catalyst descriptors are extracted from the *anode/working electrode* only, because Pt or graphite counter electrodes would otherwise be counted as the catalyst. Element and family flags (oxide, hydroxide, oxyhydroxide, LDH, phosphide, nitride, sulfide, selenide, boride, MOF) are matched on chemical formulas such as Ni_3S_2, CoNiP and Ni_2P. Substrate (foam, conductive glass, carbon), counter-electrode type and electrolyte pH, explicit or computed as 14 + log[OH^-], complete the 61 descriptors. Potential vs RHE is missing for 32 records and temperature for 63. Missing inputs are median-imputed inside each training fold, with binary missing-value indicators appended, and outcome variables are never used as inputs.

2.2. Machine Learning Models.

Decision tree (DT), support vector machine (SVM), random forest (RF), neural network (NN), gradient boosting (GB) and extra trees (ET) classifiers were compared [14]. Hyperparameters were selected from a small per-family grid under the same grouped protocol. Every score reported is the macro-averaged F1 of pooled out-of-fold predictions over 25 repeats × 5 publication-grouped folds. Accuracy, precision, recall and F1 are computed from the confusion matrix (eqs 1-4).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

2.3. *K*-Means Clustering Algorithm.

K-means partitions records into K clusters by minimising the within-cluster sum of squares in the standardised (conversion, yield) space for $K \in \{3, 4, 5, 6\}$, with 100 random initialisations (eqs 5-7). Missing outcomes were median-imputed for clustering only, and those records are flagged in all figures. A secondary binary target labels a record high-performance when min(conversion, yield) ≥ 90%.

$$d(x_i, c_j) = \sqrt{\sum_{m=1}^{M} (x_{im} - c_{jm})^2} \tag{5}$$

$$c_j = \frac{1}{|S_j|} \sum_{p_i \in S_j} p_i \tag{6}$$

$$\min \sum_{j=1}^{k} \sum_{p_i \in S_j} \|p_i - c_j\|^2 \tag{7}$$

2.4. Application of Oversampling Technology.

Random oversampling (ROS), SMOTE [10] and ADASYN [11] were applied only to training folds. SMOTE interpolates between a minority sample and one of its nearest minority neighbours (eq 8). ADASYN allocates more synthetic samples to minority points surrounded by majority neighbours (eq 9). The neighbour count is min(2, n_{min} − 1), so a class with one training record makes the sampler infeasible for that fold. Such repeats are reported, not scored, and a model/sampler pair is ranked only if at least 80% of its repeats are complete.

$$x_{\text{new}} = x_i + \lambda\,(x_{zi} - x_i), \quad \lambda \in [0, 1] \tag{8}$$

$$g_i = \frac{r_i}{\sum_i r_i} \times G \tag{9}$$

2.5. Feature Interpretability Analysis.

SHAP values [12] decompose each prediction into additive feature contributions relative to a baseline expectation (eq 10). Kernel SHAP was applied to the complete fitted pipeline, including imputation and scaling, on a publication-grouped 80/20 hold-out. For K = 3 and K = 5 it explains the best model/sampler pair of the comparison.

$$f(x) = g(x') = \phi_0 + \sum_{i=1}^{m} \phi_i x'_i \tag{10}$$

2.6. Application of the Integrated Genetic Algorithm.

NSGA-II [13] was used for multi-objective screening, following the design of ref 1 with electrochemical objectives. Predicted conversion X, FDCA yield Y and FE are maximised, while temperature T and applied potential E are minimised. These two operating burdens replace the reference article's pressure and reaction time, which this dataset does not record. Predictions come from per-target random-forest regressors (mutual-information selection of 15 descriptors, 60 trees of depth 3) that beat the median baseline in MAE for all four outputs under repeated grouped cross-validation. The decision variables use mixed encoding. Discrete descriptors are binary, and one-hot groups (electrolyte cation and anion) obey the hard constraint $\Sigma x_i = 1$. Continuous variables are min-max normalised to [0, 1]. The search used a population of 100 for 1000 generations, with binary tournament selection, blend crossover, Gaussian mutation and elitist ($\mu + \lambda$) survival by front and crowding distance. Pareto solutions were ranked with the entropy-weighted fitness of eq 11. Convergence was monitored with the spacing metric (eq 12) on fixed-scale objectives, where 0 is each objective's ideal value.

$$\text{fitness} = \omega_1 X + \omega_2 Y + \omega_3 \text{FE} - \omega_4 T - \omega_5 E \tag{11}$$

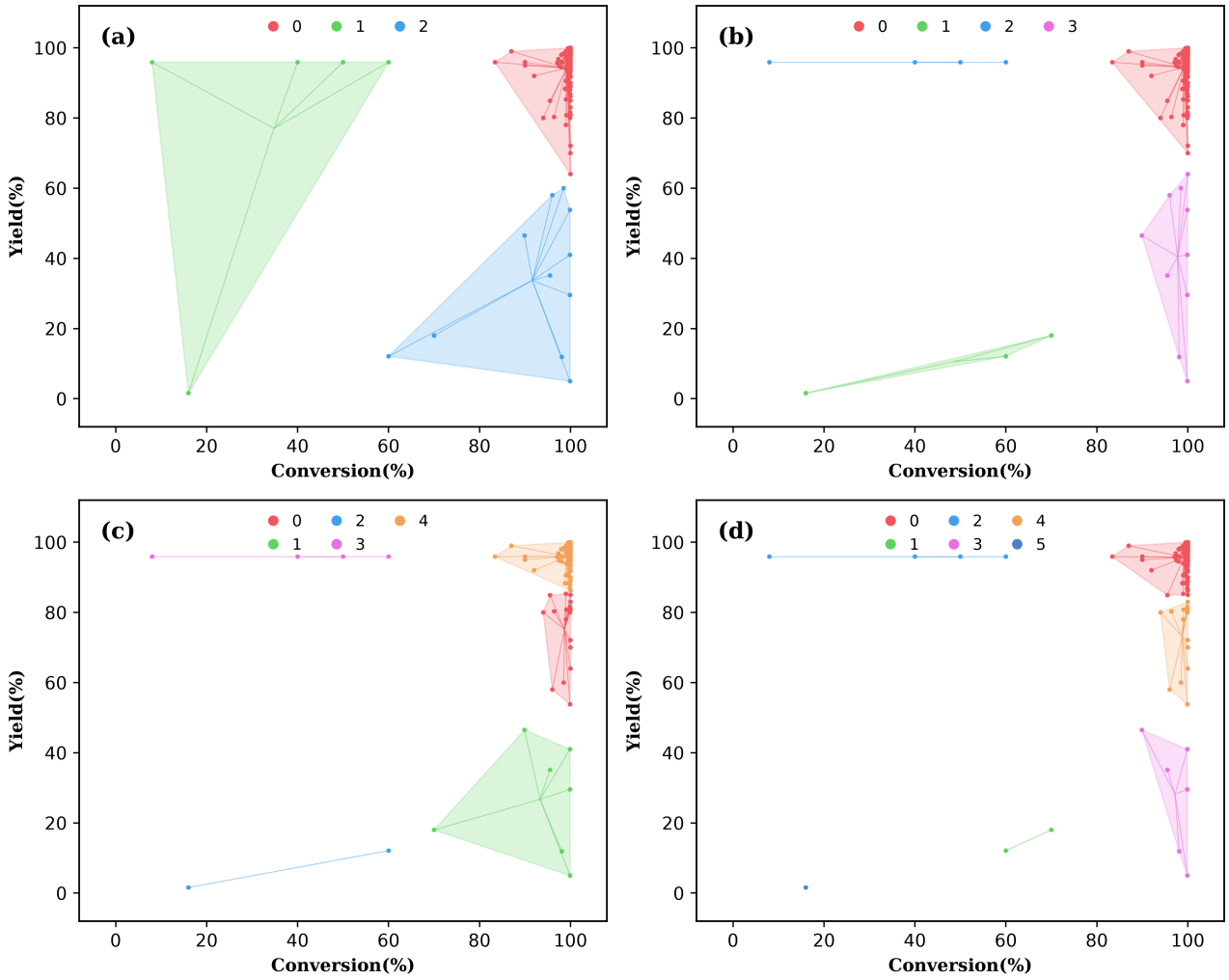
$$S = \sqrt{\frac{1}{|H|} \sum_{i=1}^{|H|} (\bar{d} - d_i)^2} \tag{12}$$

In the constrained scenario the anode is a single-metal $Ni(OH)_2$ or $Cu(OH)_2$, with all other element and family flags set to zero and derived descriptors recomputed. Potential and temperature are bounded to the range reported for real Ni/Cu-hydroxide anodes in the database (1.40-1.69 V vs RHE and 25-40 °C). Each metal is optimised separately, and the final populations are pooled and re-ranked.

3. RESULTS AND DISCUSSION

3.1. *K*-Means Clustering Results.

Figure 2 shows the clusters for K = 3-6. Unlike the thermocatalytic data of ref 1, electro-oxidation records crowd the high-conversion corner. The dominant cluster holds 138 of 154 records at K = 3. Further clusters separate the low-yield, high-conversion region, where oxidation stops at intermediates or over-oxidises, from the few low-conversion records. Higher K subdivides the high-yield region but leaves minority classes of one to four records (Table 1). That severe imbalance motivates both oversampling and the binary performance screen.



Without sampling K-means clustering results: (a) K = 3, (b) K = 4, (c) K = 5, and (d) K = 6

Figure 2. Without-sampling K-means clustering results: (a) K = 3, (b) K = 4, (c) K = 5 and (d) K = 6. Shaded polygons are cluster convex hulls, and spokes join each record to its cluster mean.

Table 1. Cluster Sizes for K = 3-6 (154 Records)

K	Cluster sizes
3	0: 138, 1: 5, 2: 11
4	0: 137, 1: 3, 2: 4, 3: 10
5	0: 17, 1: 7, 2: 2, 3: 4, 4: 124
6	0: 127, 1: 2, 2: 4, 3: 6, 4: 14, 5: 1

3.2. Performance Evaluation of the Learning Models.

Table 2 lists the best macro-F1 of each model family. Correcting the descriptors raised scores for every family. Previously the Pt cathode was counted as the catalyst in 74 records, and phosphide, nitride and sulfide anodes were never detected. Other gains came from missing-value indicators and regularised hyperparameters: the NN no longer early-stops on a 10% split of an already small fold. For the binary screen the NN reaches 0.731 and ET 0.720, against a stratified dummy baseline of 0.480. For K = 3, GB reaches 0.626. The K = 4-6 classifications remain weak (best macro-F1 0.43, 0.32 and 0.28) because their minority clusters contain only a handful of papers.

These scores are lower than the 0.75-0.85 reported in ref 1 because the evaluation is stricter. Under random splits, near-duplicate records from the same paper fall on both sides of the split and inflate the score; publication grouping prevents that. The learning curves of Figure 3 show the expected small-data behaviour: training F1 well above validation F1 for the tree models, and for SVM at K = 3 a validation score that keeps rising with the training size.

Table 2. Best Pooled Out-of-Fold Macro-F1 per Model (Best Sampler in Brackets), Before and After the Descriptor and Model Corrections

Model	Binary ≥90% before	Binary ≥90% now	K = 3 before	K = 3 now
DT	0.557	0.600 (ADASYN)	0.410	0.519 (none)
SVM	0.534	0.695 (ROS)	0.298	0.486 (none)
RF	0.586	0.678 (ROS)	0.379	0.481 (ROS)
NN	0.507	0.731 (SMOTE)	0.389	0.535 (ROS)
GB	0.575	0.661 (SMOTE)	0.421	0.626 (none)
ET	—	0.720 (none)	—	0.484 (SMOTE)

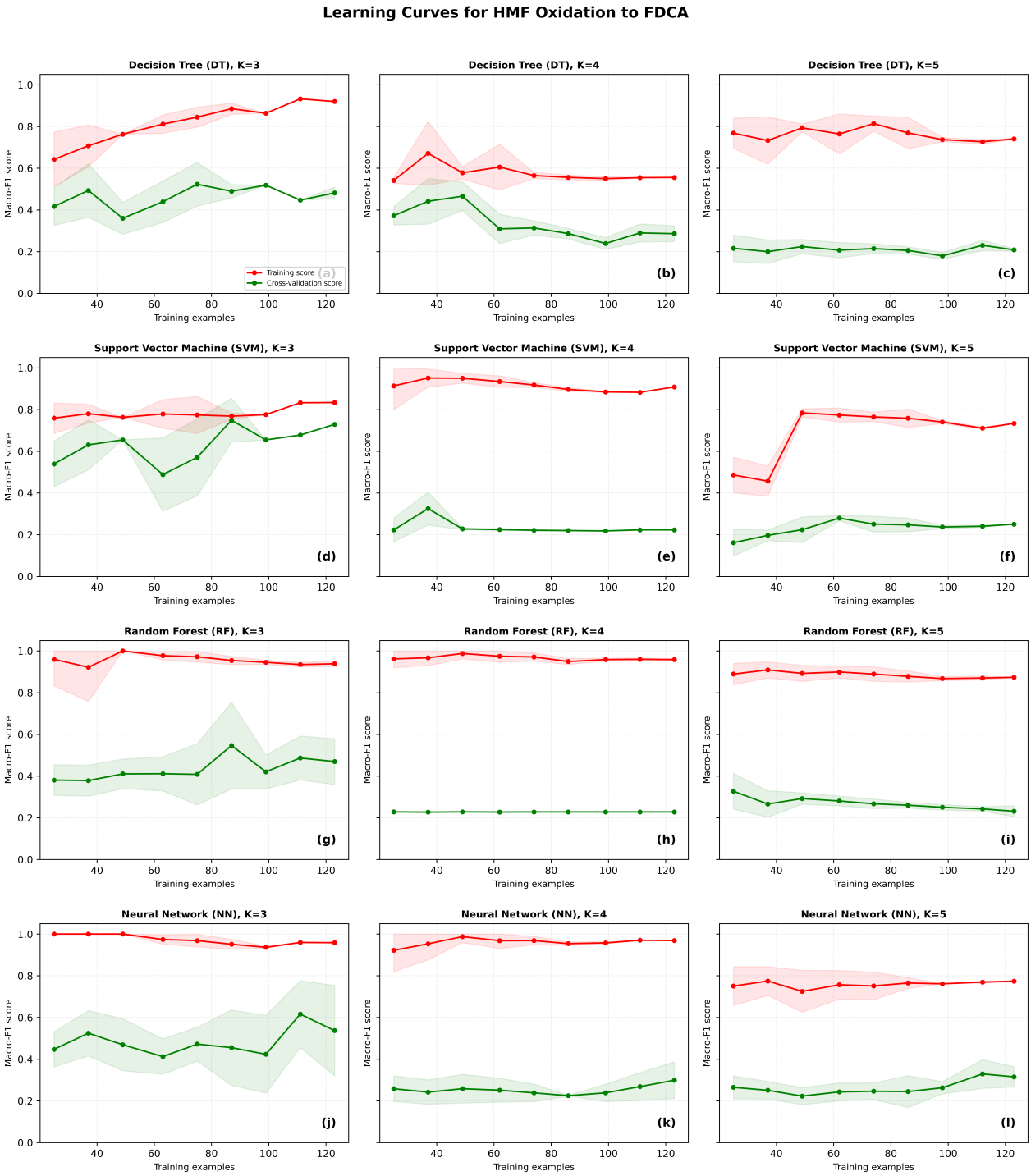


Figure 3. Learning curves of DT, SVM, RF and NN (rows) for K = 3, 4 and 5 (columns) on publication-grouped hold-outs. Red: training score; green: validation score; bands: ±1 SD over 10 repeats.

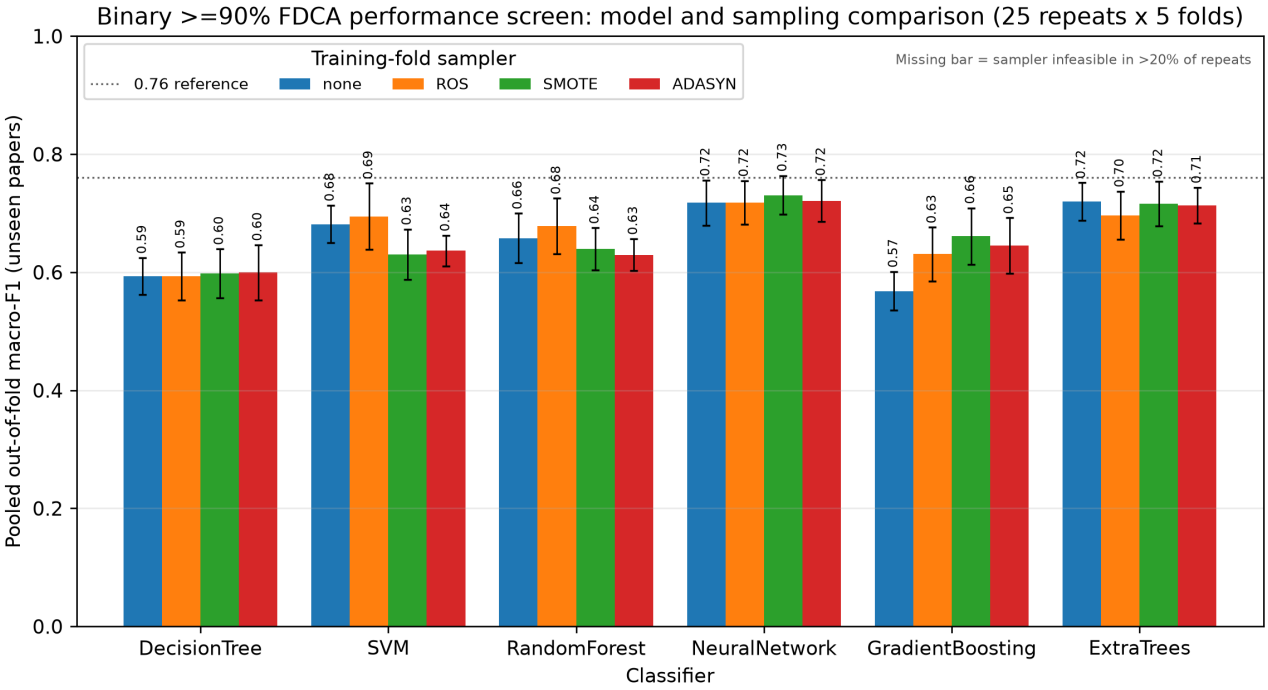


Figure 4. Model and sampling comparison for the binary ≥90% FDCA performance screen (pooled out-of-fold macro-F1, mean ± SD over 25 repeats; dotted line: 0.76 reference).

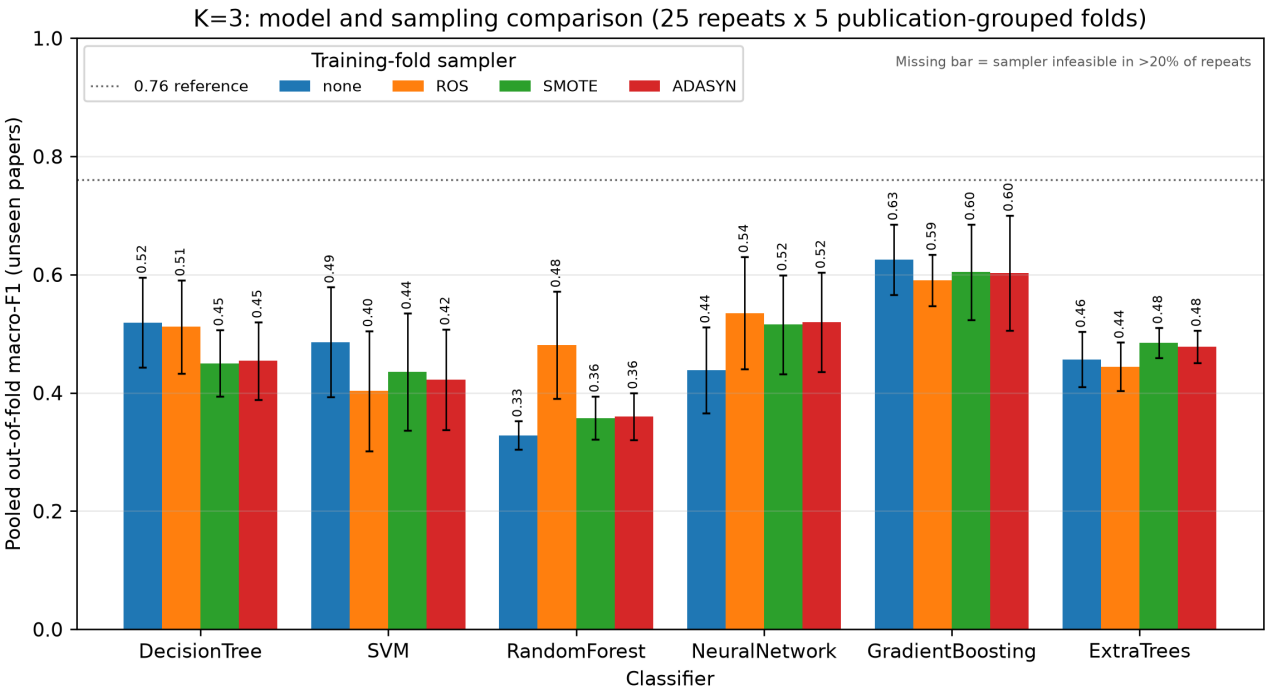


Figure 5. Model and sampling comparison for the K = 3 cluster classification.

3.3. *K*-Means Analysis after Oversampling.

Figure 6 shows the synthetic records produced by ADASYN and SMOTE in the conversion-yield plane. Interpolation fills the sparse low-yield clusters along the segments between measured records and preserves the cluster topology, as in ref 1. For K = 3 and K = 6 the interpolating samplers are infeasible because one cluster has a single measured record, so random oversampling is the only option there.

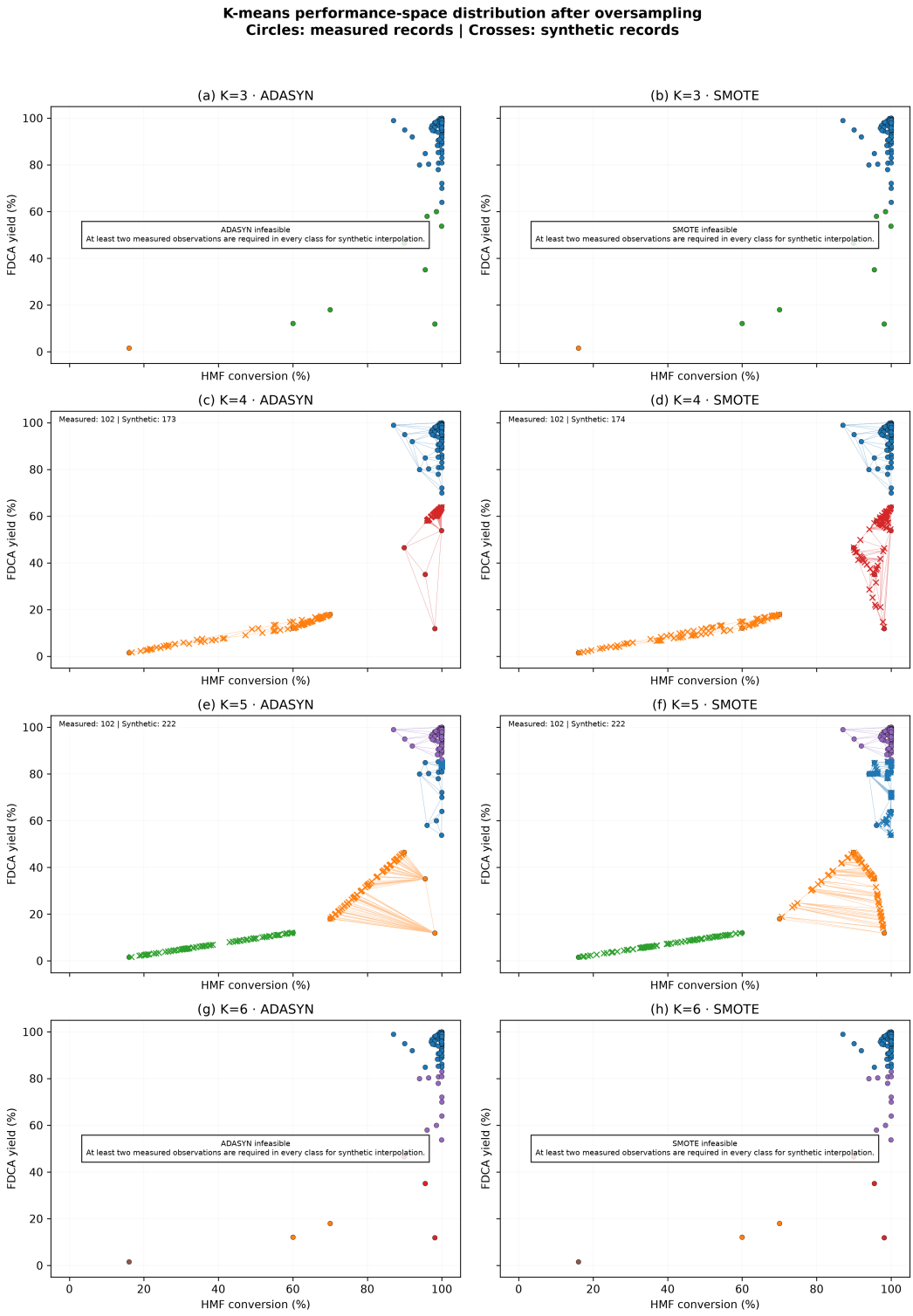


Figure 6. K-means clustering after oversampling: (a, c, e, g) ADASYN and (b, d, f, h) SMOTE for K = 3-6. Circles: measured records; crosses: synthetic records.

3.4. Analysis of Learning Curves after Oversampling.

Figure 7 and Table 3 summarise the four reference-paper models after oversampling, averaged over held-out folds (fold-level means, as in ref 1). At K = 3, RF and NN with ROS give the highest fold-averaged F1. Only combinations whose sampler was feasible in every fold are tabulated. The oversampled NN learning curves (Figure 8) show validation F1 rising with the number of training papers, which indicates the models are data-limited rather than capacity-limited.

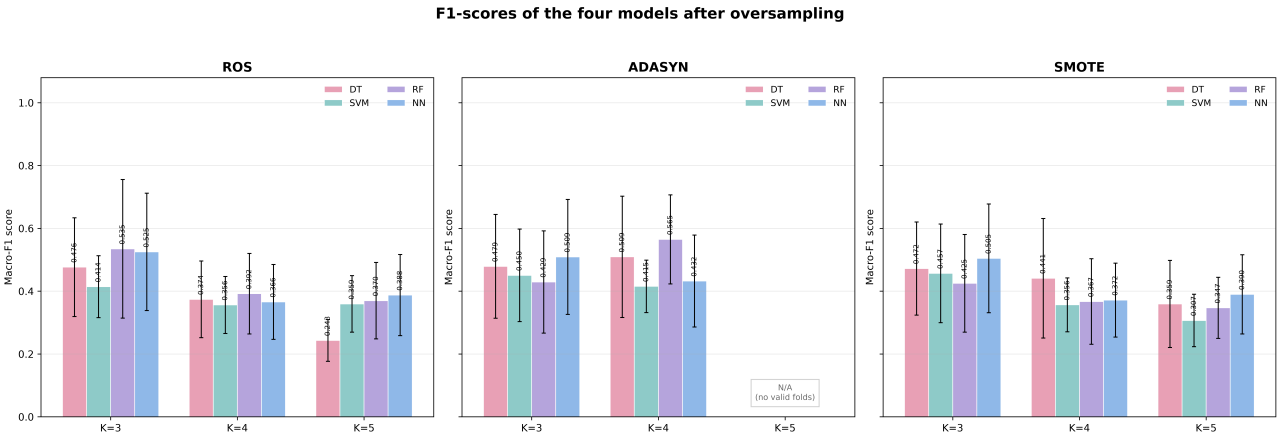


Figure 7. F1-scores of the four models after oversampling for K = 3-5: ROS, ADASYN and SMOTE (fold-level mean ± SD).

Table 3. Best Oversampled Models (All Folds Feasible), Fold-Averaged Held-Out Metrics

K	Model	Oversampling	Accuracy	Precision	Recall	F1-score
3	RF	ROS	0.8996	0.5688	0.5284	0.5348
3	NN	ROS	0.8760	0.5224	0.5498	0.5250
3	NN	ADASYN	0.8676	0.5048	0.5400	0.5090
3	NN	SMOTE	0.8671	0.5033	0.5292	0.5046
3	DT	ADASYN	0.8811	0.4674	0.5090	0.4789
3	DT	ROS	0.8477	0.4749	0.5013	0.4764
3	DT	SMOTE	0.8794	0.4642	0.4994	0.4720
3	SVM	SMOTE	0.8889	0.4469	0.4750	0.4567
3	SVM	ADASYN	0.8867	0.4386	0.4694	0.4503
4	DT	SMOTE	0.8806	0.4458	0.4522	0.4411

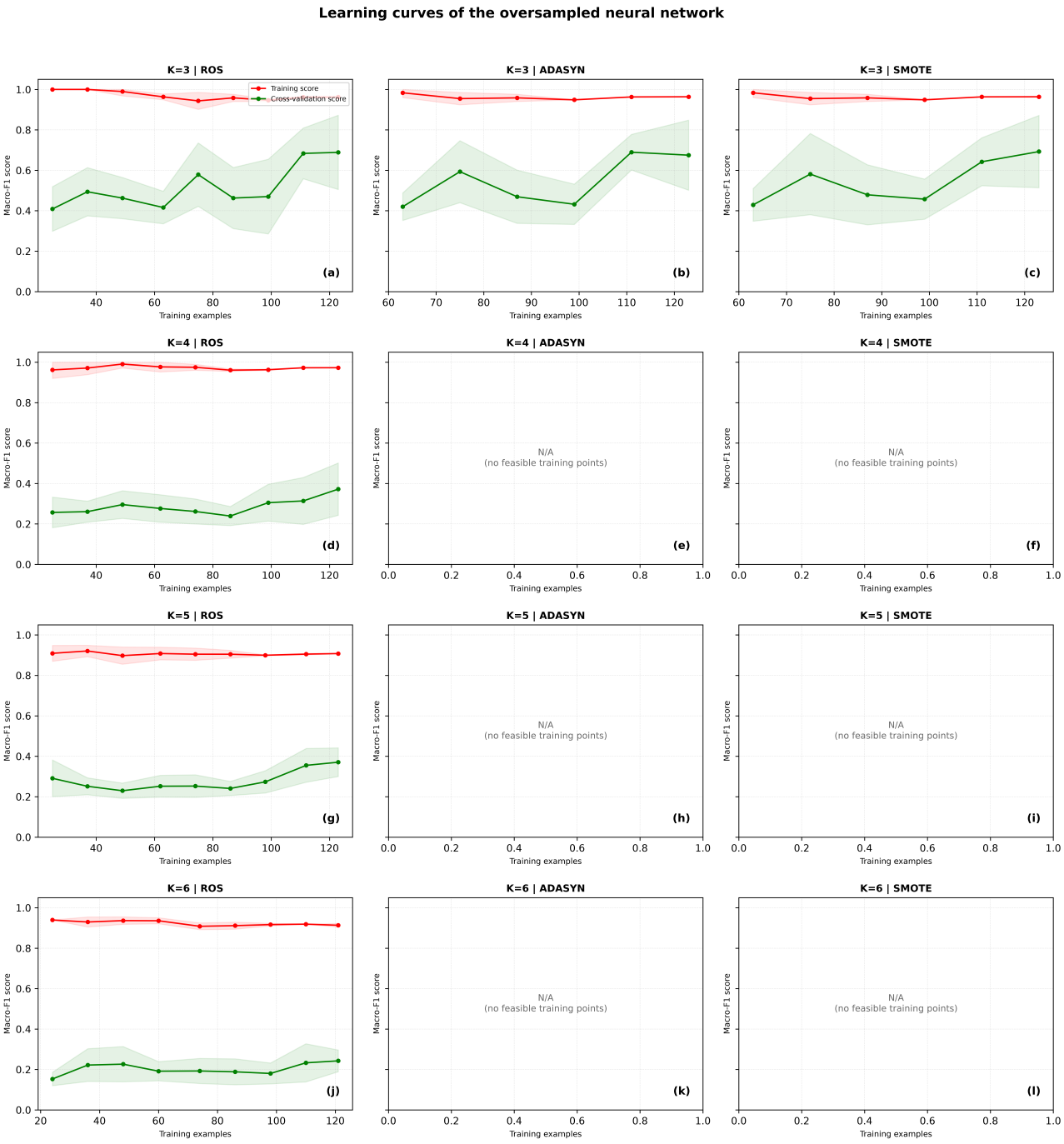


Figure 8. Learning curves of the oversampled NN model (ROS, ADASYN, SMOTE) for K = 3-6.

3.5. Results of Feature Importance Analysis.

For K = 3 the best model (GB + none) relies mainly on Catalyst_Conductive_Glass, Potential_vs_RHE_V, HMF_Concentration_mM. For K = 5 (NN + ROS) the leading features are Catalyst_Pd, Catalyst_Conductive_Glass, Divided_Cell (Figures 9 and 10). The conductive-glass flag (FTO/ITO/BDD thin films) marks a distinct group. These 16 records have a median FDCA yield of 38%, against 96% for the rest, consistent with low-loading thin films that often stop at intermediates. High-surface-area foam supports (77 records, median yield 97%) dominate the high-yield cluster. Applied potential and HMF concentration follow in importance, matching the known sensitivity of FDCA selectivity to the potential window and substrate loading. SHAP values are associational, not causal.

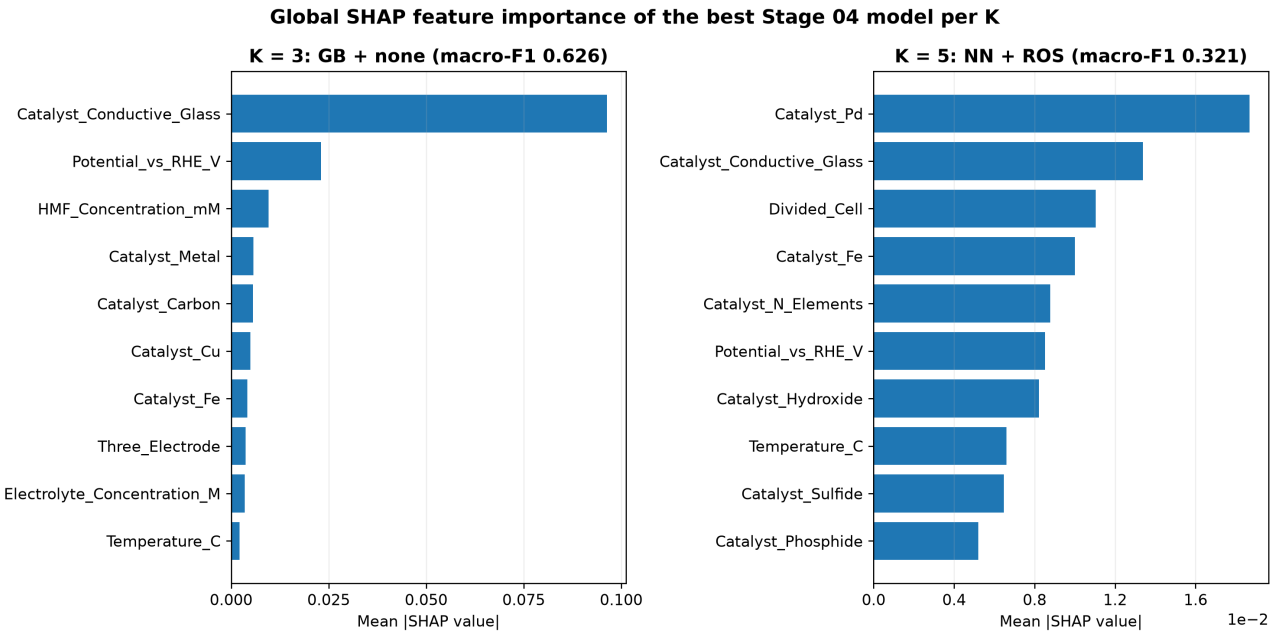


Figure 9. Global SHAP feature importance (mean |SHAP|) of the best model for K = 3 (left) and K = 5 (right).

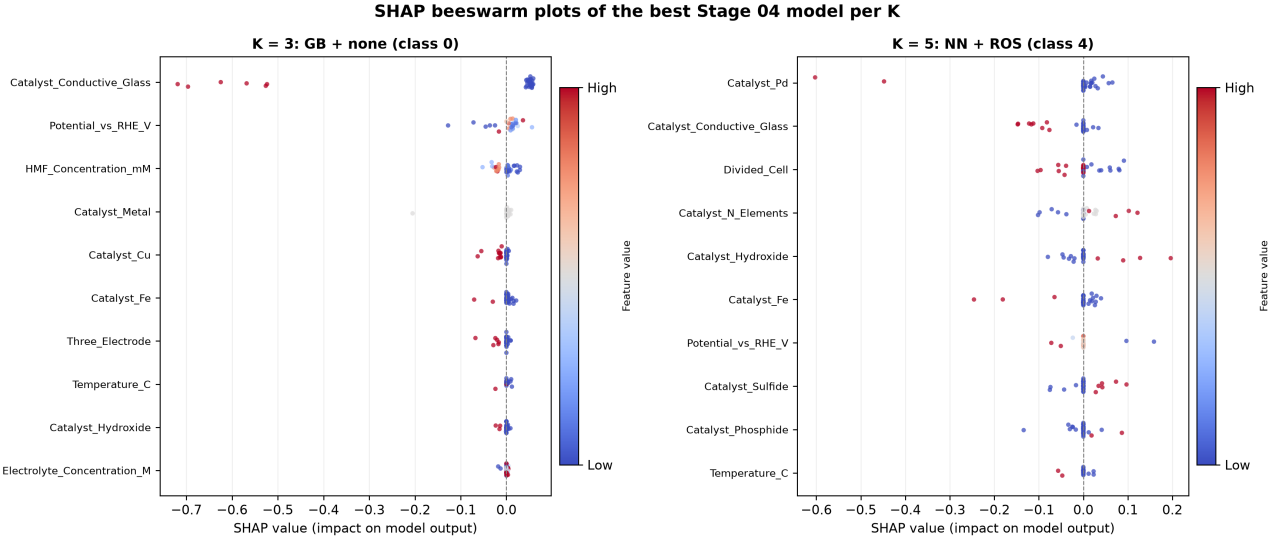


Figure 10. SHAP beeswarm plots for K = 3 and K = 5, coloured by feature value (low to high).

3.6. Application of the Integrated Genetic Algorithm in ML Modeling for Catalyst Optimization.

The validation gate passed: the per-target regressors beat the median baseline MAE for conversion (6.4 vs 7.1), yield (11.9 vs 13.2), FE (8.6 vs 9.9) and selectivity (10.0 vs 12.4). Their R^2 is close to zero, however, so predictions are shrunk toward the dataset mean and should be read as a ranking signal. The spacing metric of the first front (Figure 11) quickly settles to a low, stable level ($Ni(OH)_2$: 0.016 at generation 1, median 0.010 over the last 500 generations (max 0.026); $Cu(OH)_2$: 0.029 at generation 1, median 0.013 over the last 500 generations (max 0.027)). The population is seeded with measured records, so the initial front is already compact and there is no long transient like the one in Figure 14 of ref 1. The residual fluctuation comes from solutions that differ only in temperature or potential. Entropy weights computed on the hydroxide Pareto set were $\omega_X = 0.172$, $\omega_Y = 0.257$, $\omega_{FE} = 0.134$, $\omega_T = 0.169$ and $\omega_E = 0.268$. Potential and yield discriminate most strongly among Pareto solutions.

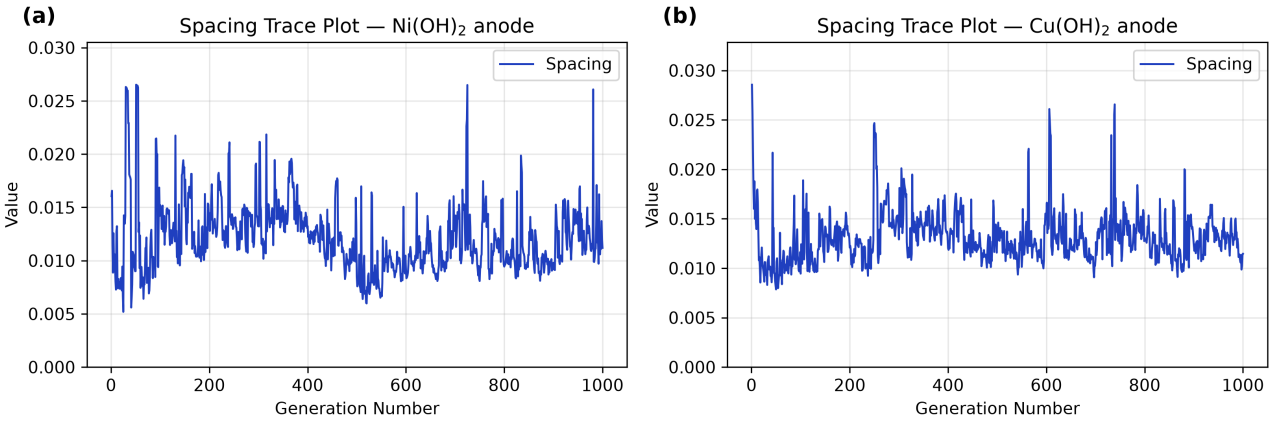


Figure 11. Spacing trace of the NSGA-II Pareto front for the (a) $Ni(OH)_2$ and (b) $Cu(OH)_2$ anode searches.

Table 4 lists the ten best solutions of the constrained search. All ten are $Ni(OH)_2$ on nickel foam in KOH (0.4-1.1 M) with ~12 mM HMF at 1.41-1.42 V vs RHE and ~25 °C. Predicted FDCA yields are 94.3-95.4% and FE 93.2-96.2%. The best $Cu(OH)_2$ solution, on Cu foam in 0.67 M KOH at 1.41 V, is predicted at 91.0% yield. It ranks lower because the database contains few high-yield $Cu(OH)_2$ records. As a control, the unconstrained search proposed Ni+Cu at 0.55 V and 9 °C, outside the region where Ni/Cu hydroxides oxidise HMF. This extrapolation by the tree regressors is why the catalyst-specific condition envelope is required.

Table 4. Top-10 NSGA-II Solutions with a $Ni(OH)_2$ or $Cu(OH)_2$ Anode, Ranked by Pareto Front and Entropy-Weighted Fitness (Model Predictions)

#	Catalyst	Support	Electrolyte	c (M)	HMF (mM)	E (V RHE)	T (°C)	X (%)	Y (%)	FE (%)	Fitness
1	$Ni(OH)_2$	foam	KOH	1.1	11.1	1.41	25.0	99.0	95.2	96.2	0.890
2	$Ni(OH)_2$	foam	KOH	0.8	12.1	1.41	25.4	99.0	95.2	96.2	0.874
3	$Ni(OH)_2$	foam	KOH	0.83	8.2	1.411	25.2	99.0	95.4	95.0	0.866
4	$Ni(OH)_2$	foam	KOH	0.84	11.9	1.412	26.1	99.2	95.4	95.1	0.855
5	$Ni(OH)_2$	foam	KOH	0.45	11.7	1.415	25.0	99.2	94.8	94.6	0.854
6	$Ni(OH)_2$	foam	KOH	0.65	8.8	1.412	25.7	99.2	94.9	94.7	0.851
7	$Ni(OH)_2$	foam	KOH	0.42	12.0	1.41	25.2	99.2	94.3	94.6	0.851
8	$Ni(OH)_2$	foam	KOH	0.69	11.0	1.416	25.0	99.2	94.9	94.7	0.850
9	$Ni(OH)_2$	foam	KOH	0.71	11.5	1.412	25.5	99.2	94.5	94.9	0.844
10	$Ni(OH)_2$	foam	KOH	0.43	11.8	1.412	25.4	99.2	94.8	93.2	0.837

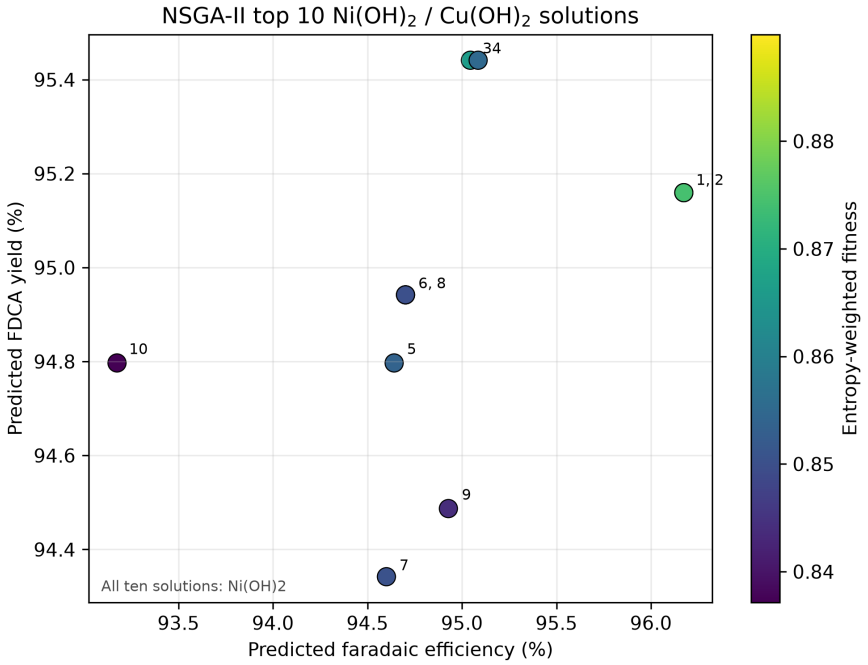


Figure 12. Predicted FDCA yield vs FE of the top-10 solutions, coloured by entropy-weighted fitness.

Comparison with experiment. The nearest database record to solution 1 is $Ni(OH)_2$/NF (1.0 M KOH | 1.45 V vs. RHE). It measured 88.3% yield and 90.6% FE, against the predicted 95.2% and 96.2%. The model is therefore optimistic by about 7 percentage points. Table 5 compares the solutions with published electrodes. A freestanding $Ni(OH)_2$/nickel-foam electrode was studied at exactly the proposed conditions (1 M KOH, 10 mM HMF, 1.45 V vs RHE) [4], and modified $Ni(OH)_2$ anodes at 1.45-1.50 V report 95-99% FDCA yield and FE [5-7]. $Cu(OH)_2$/Cu-foam reaches 98.7% yield in 1 M KOH [8]. The algorithm thus recovers the experimentally established operating window for $Ni(OH)_2$. Its absolute predictions, and the lower ranking of $Cu(OH)_2$, reflect the limits of a 154-record database.

Table 5. Published $Ni(OH)_2$ and $Cu(OH)_2$ Electrodes for HMF Electro-oxidation Used to Check the GA Solutions

Reported catalyst	Conditions	Reported performance	Ref.
$Ni(OH)_2$/Ni foam (freestanding)	1 M KOH, 10 mM HMF, 1.45 V vs RHE	Studied at the same conditions as solution 1	[4]
Vacancy-engineered $Ni(OH)_2$/CP	1.45 V vs RHE	Yield 98.5%, FE 98.5%	[5]
ZIF-67@beta-$Ni(OH)_2$/NF	1.45 V vs RHE	Yield 99.0%, FE 95.2%	[6]
$Ni(OH)_2$-CeO_2/NF	1.504 V vs RHE	Conv. 98.9%, sel. 98.8%, FE 98.4%	[7]
$Cu(OH)_2$-decorated Cu foam	1.0 M KOH, 100 mM HMF	Yield 98.7%, FE ~100%	[8]
$Cu(OH)_2$ nanowires/Cu foam	0.1 M KOH	$Cu(OH)_2$ vs CuO activity comparison	[9]

4. CONCLUSION

We transferred the ML-driven catalyst-screening workflow of ref 1 to the electrochemical oxidation of HMF to FDCA. Correcting how catalyst descriptors are extracted (anode only, formula-aware) and evaluating every model on unseen publications improved and stabilised the classifiers. The NN with SMOTE is the best high-performance screen (macro-F1 0.731), and GB is best for K = 3 (0.626). SHAP highlights the electrode architecture and the applied potential. The entropy-weighted NSGA-II search, constrained to $Ni(OH)_2$/$Cu(OH)_2$ anodes, converges on $Ni(OH)_2$ on nickel foam in about 1 M KOH with about 10 mM HMF at 1.41-1.45 V vs RHE and room temperature. That window is consistent with published electrodes reaching 95-99% FDCA yield.

Limitations. The database is small, has missing values and leans toward reported successes. The per-target regressors explain little variance ($R^2 \approx 0$), so predicted performance is optimistic by about seven points relative to the closest measured record. Hyperparameters were chosen on the same records. The within-paper label-permutation test is not significant ($p = 0.47$), which shows that most signal separates papers rather than conditions within a paper. GA solutions are hypotheses for experimental validation, not optimal catalysts.

DATA AND CODE AVAILABILITY

All figures and tables were generated by *run_all.py* (stages 01-10) from *data/raw*. Numerical results are in *results/*, including *classification_summary.csv*, *nsga2_top10_ni_cu_hydroxide.csv* and *FINAL_RESEARCH_REPORT.md*.

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